CA-I-F-22

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Architecture

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the term Computer Architecture. Explain IAS computer architecture with detailed diagram.
   b) Define addressing mode. Explain different addressing modes with appropriate figures and example of each. 7
2. a) Define RTL. How data can be transferred between registers. Explain. 8
   b) Define Instruction cycle. What are the different states in an instruction cycle? Also explain fetch cycle, decode cycle, execute cycle and indirect cycle. 8
3. a) Draw the flowchart for Booth's Algorithm. Use the algorithm for binary multiplication of (-12) and (-9). 8
   b) Explain single and two address field sequencing techniques in micro programmed control unit. 7
4. a) Differentiate between Hardwired and Micro-programmed control unit. Explain the logic of Hardwired unit. 7
   b) What is memory hierarchy? Explain with reason why do we need multilevel memory hierarchy? 7
5. a) Explain various communication techniques for I/O devices. 8
   b) Differentiate between various mapping techniques when main memory data is to be mapped into cache memory. 7
6. a) What is power efficient processor? Explain dual core processor with respect to quad core processor. 8
   b) Describe in brief about Flynn's Taxonomy. Also explain how parallelism can be experienced in uniprocessor systems. 7
7. Write short notes on: (Any two) 2×5
   a) RISC vs. CISC
   b) Vector processors and Array processors
   c) BCD Adder

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